

Scope: A Cognitive Model of Thought Generation Pathways to Prevent Single-Interpretation Fixation of Misunderstanding and Division

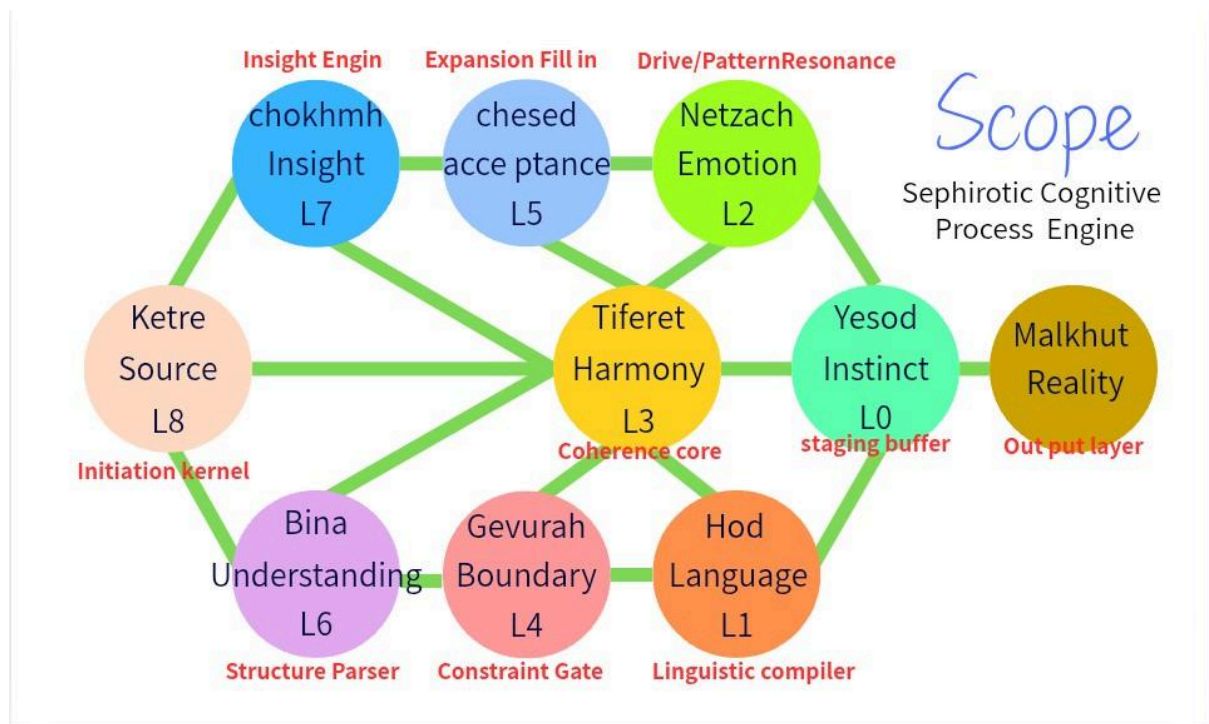


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0. Overview

0-1. Scope is not a single-function tool

Scope cannot be reduced to a single function or purpose.

Scope is a “map” for observing and extracting the process of human thought generation, and structurally inferring the pathways through which that thought is produced.

This map does not evaluate the correctness of thoughts or their conclusions. Instead, it functions as a foundation for handling “how a thought was generated” as multiple hypotheses.

Scope does not directly execute or control actions. However, it functions as a decision-making foundation that shapes how thoughts and responses are handled.

Due to this nature, Scope is not limited to a specific application. It can be referenced as a common structure across multiple domains, including analysis, dialogue understanding, self-observation, and response design.

Therefore, Scope is not a single-purpose tool for solving a specific problem, but a foundational framework for handling the structure of thought generation.

0-2. Focus of this document (Construction of a “State of Inference”)

This document focuses on the fixation of single interpretations in LLMs and the resulting problem of misaligned interventions.

Current LLMs tend to generate the most plausible single interpretation for a given input and produce responses based on it. This tendency can lead to the fixation of misunderstandings, excessive agreement, or inappropriate rejection, which in turn may amplify division and conflict.

Nonviolent Communication (NVC) is an effective method for stabilizing dialogue. However, when underlying assumptions or interpretations significantly diverge, or when defensive or aggressive reactions are strongly present, adjusting the form of communication alone may not be sufficient.

This document addresses such situations by constructing a “state of inference” in which the pathways of thought generation can be handled as multiple hypotheses.

This state does not fix interpretation to a single conclusion, but maintains multiple possible pathways simultaneously.

Scope is positioned as the structural foundation that enables this state of inference.

Although this framework is designed for human–AI interaction, it can also be applied to human–human interaction when similar structural conditions are present.

However, its range of applicability is currently based on limited observations, and further validation is required for generalization.

0-3. Relationship with COS

COS (Cognitive Operating System) is an internal structure responsible for processing premises and context in response to input, and for detecting misalignment in interpretation. The interpretation handled here includes not only linguistic meaning, but also context, nuance, and tone.

Through this, it becomes possible to clarify the process by which a particular interpretation is reached based on specific premises and contextual conditions. In human-to-human interaction, this type of processing is often performed unconsciously. However, in LLMs, this interpretive process is unstable, and mismatches in premises tend to surface more easily.

Detailed structures and processes of COS are covered in a separate document. This document focuses on Scope.

For example, even the same word can have different meanings depending on underlying premises. The term “chips” may refer to potato chips in one context, and to French fries in another. Interpretation is dependent on the assumed context.

However, the issue extends beyond differences in meaning. For instance, when someone asks, “I want to travel alone— is that possible?”, an appropriate response cannot be determined without clarifying the underlying premises. Whether the speaker is an adult or a minor significantly changes what kind of response is appropriate. Current LLMs may fail to verify such premises and instead generate a generalized response.

In this way, while COS can identify mismatches in premises, it does not by itself determine why a statement was generated or how one should respond to it.

For example, when a minor expresses a desire to travel alone, there may be multiple possible underlying reasons. It could be simple curiosity or interest, or it may be related to family circumstances or personal distress. Responding based on a single interpretation at this stage carries the risk of inappropriate or unsafe outcomes.

Scope addresses this limitation. Scope structurally infers the pathways through which a thought was generated and maintains multiple hypotheses. This enables the system to consider the generative process behind an utterance, rather than fixing on a single

interpretation. As a result, it becomes possible to determine whether clarification, exploration, or other forms of engagement are appropriate.

Therefore, COS and Scope are not opposing systems, but complementary layers operating at different levels. COS handles the processing of premises and interpretation, while Scope addresses the underlying generative pathways. Together, they enable understanding and decision-making that do not rely on a single interpretation.

1. Case Studies

1-1. Conflict Arising from Mismatched Social Expectations

Consider a situation in which a person feels that “not greeting someone is rude” and develops dissatisfaction toward another individual. For this person, greeting is a basic social behavior, and failing to do so is interpreted as disrespectful based on their underlying premise.

However, the other party does not necessarily share the same premise. There may be multiple possible reasons why a greeting did not occur, such as not noticing the other person, being distracted, or being occupied with another situation.

For example, it is possible that the person did not recognize the other’s presence, or that perceptual or attentional factors prevented the opportunity for greeting from arising in the first place. Such factors are difficult to observe externally and are often treated as intentional disregard.

When the reason behind an action is not explicitly clarified, interpretation tends to converge into a single evaluation, such as “a person who does not greet is rude.” As this evaluation becomes fixed, negative emotions and responses may be justified.

In such situations, when a third party—especially an LLM—receives this interaction, the interpretation may become further simplified. If only the perspective of the person who felt ignored is provided as input, the premise that “the other person is rude” is likely to be accepted as given, leading to responses that reinforce this view through empathy or advice.

At this point, the possibility of alternative explanations for the lack of greeting is not considered, and responses based on a single interpretation are generated. As a result, rather than deepening understanding, one-sided recognition becomes reinforced and fixed.

A similar structure can also occur in human interactions. When one individual interprets an event as “being ignored” and shares this interpretation with others, it may be accepted as fact and gain agreement from the group.

In this process, alternative interpretations that could have existed are not examined, and the single premise of “being ignored” becomes collectively shared. This leads to a unidirectional fixation of perception within the relationship, potentially justifying negative responses toward the targeted individual.

Furthermore, when actions begin to change based on this shared premise, the initial interpretation may become reinforced as an actual outcome. In this way, a single interpretation is reproduced at the group level and becomes fixed.

In addition, when LLM responses are introduced into such situations, their influence can further amplify the effect. For instance, if an LLM provides an empathetic response to the perception of “being ignored,” that response may be treated as a third-party judgment and used to justify the claim that “objectively, the other person is at fault.”

In this case, although the LLM’s response is based only on limited input and a single interpretation, it may function as if it were independent evidence or evaluation. As a result, the opportunity to revise or reconsider the interpretation is lost, and the single perspective becomes more rigidly fixed.

In such situations, the issue lies not in the behavior itself, but in the premises on which the behavior was interpreted, and in how unobservable factors were supplemented and treated as intention.

1-2. Cases of Aggressive and Confrontational Language

Consider a situation in which an individual makes strong, definitive claims such as “I have superior abilities, so others should obey me,” or uses language that directly denies or devalues others. On the surface, such expressions appear as aggressive or confrontational language.

In conventional dialogue approaches, responses often attempt to adjust the expression itself, such as suggesting that the wording is too strong or that a more considerate tone should be used. However, in many cases, such expressions are not merely a matter of wording, but are strongly dependent on underlying premises and internal states.

For example, when an individual perceives that they are not being properly evaluated or understood by others, this perception may be accompanied by the judgment that such treatment is unfair, along with the premise that “there is nothing wrong with me.” In order to maintain this premise, statements may be generated in a way that denies or devalues others.

In such cases, the expression is not simply an attack, but has a defensive structure aimed at maintaining internal consistency. Therefore, interventions that focus only on softening the expression or providing empathy are unlikely to change the structure of the statement unless the underlying premise itself is addressed.

Similarly, negative statements toward specific groups or attributes cannot always be treated as simple misinformation. When such statements are grounded in strongly fixed premises or value systems, directly rejecting them may threaten the individual's internal consistency, potentially leading to stronger resistance or escalation of conflict.

Under these conditions, directly intervening at the level of the expressed language often fails to sustain dialogue. This is because the core issue lies not in the language itself, but in the structure that generates it.

Furthermore, when a third party—particularly an LLM—receives such input, the responses tend to polarize. On one side, the response may reject the statement and intensify conflict; on the other, partial agreement or accommodation may reinforce the underlying premise.

In both cases, responses are based on a single interpretation, and the structural background of the statement is not considered, leading to the fixation of misunderstanding and conflict.

In addition, when such fixed, confrontational premises are present, LLM responses can impact user experience (UX). For example, if an LLM responds with one-sided rejection or presents generalized, neutral statements, users may feel that their perspective or state has not been understood.

At this point, regardless of whether the response is objectively correct, the perception of “not being understood” can reduce trust and continuity in the interaction. As a result, users may disengage from the dialogue or intensify their expressions, further reinforcing conflict.

Thus, responding to aggressive or confrontational language based on a single interpretation not only fails to resolve the issue, but can also directly lead to degraded UX and breakdown of dialogue.

In such cases, instead of evaluating the statement at face value, it is necessary to treat “the underlying premises and conditions from which the statement was generated” as multiple hypotheses.

2. Problems When Left Unaddressed

2-1. Fixation of Misunderstanding

As shown in the previous section, mismatches in interpretation are initially only one of several possible explanations. However, when they are left unexamined, they gradually become fixed as “facts.”

For example, the perception that “I was not greeted” could originally have multiple possible interpretations. However, if the underlying factors are not examined and the same interpretation is repeatedly referenced, it begins to be treated as an unquestioned premise.

At this point, the interpretation shifts from a hypothesis to a fixed recognition, and any new information is incorporated only in ways that reinforce that recognition. As a result, opportunities for revision are lost.

This process can unfold as follows:

An individual A perceives that they were “ignored” and inputs this into an LLM. The LLM generates a response such as “that’s terrible,” based on the given input. A then shares this response with others, reinforcing their interpretation by saying, “the LLM also said it was wrong.”

If another individual B inputs a similar statement into an LLM, a similar response is likely to be generated. At this point, these responses may be perceived as multiple independent judgments.

In reality, however, these outputs are generated from the same underlying premise and are therefore identical in structure. What is being reproduced is not diversity of judgment, but consistency of output resulting from shared premises.

As a result, a misunderstanding that could have been corrected becomes fixed as a seemingly validated recognition, reinforced by apparent reproducibility.

This strengthened recognition can then influence behavior toward the target individual, potentially justifying actions such as ignoring or excluding them, and leading to deterioration in relationships.

2-2. Deterioration of Relationships

When a misunderstanding becomes fixed, changes begin to occur in the relationship between the involved parties. As a single interpretation is treated as definitive, perceptions and expectations toward the other person shift, and these changes are reflected in behavior.

For example, if the interpretation that “someone did not greet me” becomes fixed, the person in question may be perceived as “rude,” and interactions begin to align with that perception. This can lead to behaviors such as ignoring the individual or creating distance.

On the other hand, the person who did not greet may notice the change in behavior and attempt to explain or clarify their perspective. However, when the underlying premise is already fixed, such explanations are often interpreted as “excuses” or “justifications,” and fail to correct the misunderstanding.

Furthermore, if both parties consult an LLM from their respective perspectives, responses will be generated based on each input. As a result, both individuals may receive reinforcement that “their own interpretation is correct.” This leads to a state in which each side denies the other’s perspective, further eliminating the possibility of meaningful dialogue.

In this way, when a misunderstanding remains uncorrected, what initially was a simple difference in interpretation develops into a deterioration of the relationship through changes in behavior. Moreover, the deteriorated relationship itself becomes new evidence that reinforces the original misunderstanding, forming a self-reinforcing cycle.

At this stage, simply providing additional information is often insufficient to repair the relationship. Interventions that challenge the underlying premise may be required, but such attempts are likely to encounter strong resistance.

Although this situation may appear difficult to resolve, the core issue lies not in the dialogue itself, but in identifying which premise became fixed and at what point the opportunity for correction was lost.

2-3. Acceleration of Division

As misunderstandings become fixed and relationships deteriorate, their impact extends beyond individual interactions and scales to the group level.

When a particular interpretation is shared, individuals who hold similar views begin to align, reinforcing specific perspectives within the group. At this stage, interpretations that were originally dependent on individual circumstances start to be treated as “general tendencies” or “attributes.”

For example, a perception that a specific person is “rude” may expand into a broader assumption such as “people of that type are like that,” transforming an individual case into a collective evaluation.

In this process, alternative interpretations and exceptions are gradually excluded, and the shared premises within the group become increasingly uniform. As a result, the capacity to accept different perspectives diminishes, and cognitive bias is amplified.

Furthermore, when LLMs become involved in this dynamic, inputs that align with the already established premises tend to increase, leading to repeated generation of similar responses. This creates the appearance that the group’s perception is externally validated, further accelerating division.

Such formation of group-level premises can be inappropriately applied to individuals with specific circumstances. In particular, when individuals possess factors, conditions, or traits that are difficult to observe externally and that influence interpretation, these factors may be overlooked. As a result, such individuals are not treated as exceptions but rather as deviations.

Consequently, behaviors that should be interpreted on an individual basis are instead judged uniformly based on group-level assumptions, leading to imbalanced evaluations and responses.

In this way, individual interpretations evolve into group-level premises through processes of sharing and reproduction, ultimately reinforcing opposing structures and accelerating division.

2-4. Polarization of Responses (Agreement / Rejection)

As misunderstandings become fixed, relationships deteriorate, and these dynamics are shared at the group level, the information being input itself becomes strongly dependent on specific premises. Under such conditions, a tendency for LLM responses to polarize can be observed.

On one side, responses express empathy or agreement with the given premise. These responses reinforce the user's existing interpretation, further strengthening already fixed perceptions.

On the other side, responses may reject the input and rely on general norms or widely accepted values. In such cases, users are more likely to feel that their perspective or situation has not been understood, which can lead to disengagement or escalation.

In both cases, responses are generated based on a single assumed premise, without considering alternative possibilities or the underlying generative process. As a result, flexible and context-sensitive responses become difficult to produce.

Furthermore, such input bias is not limited to intentional manipulation but often emerges naturally from human cognitive tendencies. In many cases, individuals do not fully present information that contradicts or weakens their own position, instead prioritizing inputs that reinforce their current interpretation.

As a result, LLMs generate outputs based on limited premises, and those outputs are consistently reproduced in alignment with those premises. In this way, biased inputs and corresponding outputs form a reinforcing loop, making correction increasingly difficult.

This polarization is not a problem inherent to LLMs themselves, but rather a phenomenon that arises from the constraints embedded in the input's premises and context. When premises are fixed, outputs are likewise constrained, leading to convergence toward simplified responses of either "agreement" or "rejection."

Thus, the cycle of single-interpretation-based inputs and outputs not only hinders the correction of misunderstandings but also contributes to the persistence and amplification of conflict and division.

3. Scale Issues (Societal Impact)

3-1. Reproduction of Single Interpretations by LLMs

LLMs generate responses based on input. When inputs containing similar premises are repeated, similar responses are stably generated. This characteristic leads to a state where specific interpretations and perceptions are repeatedly produced.

In this process, what is generated is not diverse perspectives, but a single interpretation based on the same premise. However, when such outputs are repeatedly presented across multiple contexts and users, they may come to be treated as if they were generally valid or widely accepted understandings.

Furthermore, because LLM outputs tend to be consistently reproducible, they are more likely to be perceived as reliable and objective. This characteristic accelerates the reproduction of single interpretations.

Such a structure goes beyond a mere tendency of responses and begins to influence real-world decision-making and interactions. When responses based on the same premise are repeatedly generated with consistency, incorrect interpretations and judgments may also be stably reproduced.

In addition, even a single output may be repeatedly referenced across different contexts and users, leading it to be perceived as support from multiple others. As a result, an interpretation based on a single premise may be treated as a general consensus or a collectively validated view, reinforcing individual judgments.

Moreover, in such situations, outputs may be treated as external judgments, separating responsibility from the individual and making stronger or more extreme claims more likely to be adopted.

As a result, interpretations dependent on specific premises are widely shared, forming a structure in which similar perceptions are repeatedly generated and reinforced without considering individual contexts or conditions.

3-2. Structural Amplification of Division

LLMs generate responses that align with the input premises. When groups with different premises repeatedly input information aligned with their own perspectives, responses are generated that reinforce each group's existing beliefs.

Through this repeated process, shared recognition within each group is strengthened, increasing the likelihood that members will perceive their own understanding as correct. Meanwhile, perspectives from groups with different premises are less likely to be referenced, and remain unadjusted.

Additionally, since LLMs are not designed to integrate across conflicting premises, their ability to bridge opposing viewpoints is limited. As a result, each position continues to be reinforced internally, amplifying the overall structure of division.

Furthermore, when responses are generated that are consistent with each group's premises, both sides become more confident in their own interpretations, reinforcing division without mechanisms for reconciliation.

3-3. Loss and Diffusion of Context

In LLM-mediated information exchange, context is often already missing at the input stage, and such missing context may not be reconstructed, instead being treated as a given premise.

Responses generated under these conditions are based on context-deficient information, and may be reused without adequately reflecting background or situational factors.

As such responses are further shared and reused, context-free information continues to spread, forming a structure in which simplified interpretations are widely shared without consideration of specific conditions or contexts.

However, this phenomenon does not occur in all cases, but tends to emerge under specific conditions.

3-4. Emergence of Uniform Responses (Templating)

When inputs based on specific premises or structures are repeatedly provided, the corresponding responses are also likely to be generated in similar forms.

As a result, instead of producing diverse responses tailored to individual situations, outputs tend to converge into consistent patterns. While these responses may appear appropriate at first glance, they do not necessarily reflect differences in context or background.

Furthermore, as such templated responses are shared and reused, the structure in which similar inputs produce similar outputs becomes reinforced.

In this way, responses lose their specificity and converge into standardized patterns based on a single premise.

Moreover, when such context-insensitive responses are repeatedly generated, they may create a perception of not being truly understood, even if they are formally appropriate, leading to a decline in dialogue continuity and trust.

4. Why It Occurs (Structure of Misalignment)

4-1. Premise Mismatch (D2)

The misunderstandings and divisions described so far are not accidental phenomena dependent on specific situations, but arise from the structure of cognitive processing itself. One of the core factors is premise mismatch.

D2 refers to the point at which a mismatch in premises is detected during processing.

Premise mismatch refers to a state in which the initial premise (A1) used for processing does not align. Even when the same input or event is being handled, differences in the interpretive foundation lead to divergence in subsequent processing pathways.

Furthermore, this misalignment is not limited to premises alone, but can occur simultaneously across multiple processing layers. As a result, mismatches may exist concurrently across different layers.

Such misalignment may occur at a single layer or across multiple layers simultaneously. While single-layer mismatches are relatively easy to detect, when multiple layers are misaligned at once, they reinforce each other, making detection more difficult.

Ideally, such mismatches should be detected through D2, triggering re-evaluation and adjustment. However, in practice, processing often continues without such detection, allowing the misalignment to persist into subsequent stages.

Even when a mismatch occurs, it is not always explicitly recognized. A sense of inconsistency may be absorbed into interpretation without being identified as a premise mismatch.

Moreover, even when recognized, mismatches do not necessarily lead to correction. Factors such as maintaining conversational flow or internal consistency may result in the mismatch being ignored.

The critical issue is not the existence of mismatch itself, but the continuation of processing without detection or adjustment. Interpretations and decisions generated under misaligned premises can have lasting effects on subsequent cognition and behavior.

Thus, premise mismatch is not a single misunderstanding, but a structural misalignment that can occur at multiple stages of processing, compounding into larger distortions of understanding.

4-2. Contextual Misalignment

Even when premises are aligned, interpretations may still diverge. One of the causes is contextual misalignment.

Context refers to the situational and relational conditions surrounding an utterance or piece of information, including temporal sequence, relationships, cultural background, and implicit assumptions. These elements are often not explicitly stated and are reconstructed differently by each interpreter.

As a result, even when the same expressions and premises are used, meaning can change depending on the context in which they are interpreted.

Additionally, context is often omitted during transmission and may be lost over time. Even if understanding was aligned at the moment of expression, misalignment may occur during subsequent communication.

Similarly, in LLMs, when contextual information in the input is limited, it may not be reconstructed and is instead inferred based on existing patterns, potentially leading to interpretations based on incorrect contexts.

Thus, contextual misalignment differs from premise mismatch in that it generates divergent interpretations from the same starting point, making it a key structural factor in misinterpretation.

In some cases, contextual misalignment may lead to interpretations that significantly deviate from the speaker's intent, affecting real-world decisions and actions.

4-3. Interpretation Layer Differences (B1–B3)

Even when both premises and context are shared, interpretations may still differ. One factor is the difference in interpretation layers.

Interpretation is not a single process, but occurs across multiple layers (e.g., B1–B3), corresponding to different levels such as literal meaning, contextual meaning, and inferred intent.

Even with the same utterance, interpretation varies depending on which layer is primarily referenced. As a result, differences in interpretation may arise even under identical premises and contexts.

Moreover, the selection of interpretation layers is often unconscious, meaning individuals may not be aware of the layer they are using, leading to unnoticed misalignment.

Thus, interpretation layer differences are a structural factor that can produce divergent interpretations even under the same conditions.

For an overview of the internal processing structure (COS), refer to the appendix. Detailed definitions and specifications of COS are provided in separate materials.

4-4. Convergence to a Single Path

Premise mismatch, contextual misalignment, and interpretation layer differences all create conditions where multiple interpretive pathways could exist simultaneously.

However, in practice, both human cognition and LLM processing tend to converge toward a single interpretation.

In humans, cognitive load reduction and the need for internal consistency favor selecting one interpretation. Similarly, LLMs generate a single output based on probabilistic optimization rather than maintaining multiple interpretations.

Importantly, this convergence does not imply that alternative pathways do not exist, but rather that they are integrated and compressed during processing, becoming effectively invisible.

In the early stages of cognition, multiple pathways may arise based on different premises, contexts, and interpretation layers. However, as processing progresses, these pathways are integrated to maintain coherence and finalize decisions. This integration occurs primarily at the stages of coherence generation (L3) and boundary fixation (L4), and unselected pathways are not retained explicitly.

While this convergence is efficient, it also leads to situations where alternative possibilities are not sufficiently considered. As a result, mismatches in premise or context may remain uncorrected and become fixed.

Therefore, convergence to a single path is not the absence of alternative interpretations, but the structural tendency for them to be compressed and rendered invisible during processing. This invisibility contributes to the persistence of misunderstanding and division.

Addressing these misalignments cannot be achieved through a single method.

This document introduces three approaches operating at different layers: external dialogue adjustment (NVC), internal processing structure (COS), and inference of thought generation pathways (Scope).

Each of these approaches serves a distinct role and operates on the same problem from different layers. In the following sections, these approaches will be examined in sequence, ultimately clarifying the structural role that Scope occupies.

5. Positioning of NVC (External Dialogue)

5-1. Observation, Feeling, Need, Request

Nonviolent Communication (NVC) is a framework that decomposes expression in dialogue into four elements: Observation, Feeling, Need, and Request. Through this structure, it aims to organize information transfer and reduce misunderstanding and conflict.

In this context, “observation” is not merely a description of facts; it is an operation that separates evaluation, interpretation, and inference. NVC focuses not on what is communicated, but on how it is communicated.

Humans tend to unconsciously supplement meaning from observed facts and generate interpretations through associative processes. For example, an observation such as “the tone was harsh” may naturally connect to an inference like “I am disliked,” showing how facts and interpretations are easily linked in a continuous chain.

However, this connection does not necessarily represent a confirmed causal relationship; it is merely a chain of interpretations. This chain becomes directly tied to emotional responses, leading to reactions such as anxiety or discomfort.

NVC separates this process, distinguishing observation from inference, thereby reducing the amplification of misunderstandings caused by unnecessary associations.

5-2. Scope of Effectiveness (Stabilization of Communication)

NVC is effective as a method for stabilizing information transfer in dialogue. By decomposing expression into the structure of Observation, Feeling, Need, and Request, it organizes the mixture of interpretation and emotion contained in statements, thereby facilitating mutual understanding.

In particular, the separation of observation and inference suppresses unconscious associative expansion and meaning completion, helping prevent the amplification of unnecessary misunderstandings in communication.

NVC demonstrates its effectiveness when a certain level of dialogue is already established. It works under conditions where the other party has the capacity to receive input, where a minimal shared basis of understanding exists, and where adjustment of output is possible. In such contexts, it contributes to reducing misunderstandings and mitigating conflict.

Additionally, separating observation and inference can relativize interpretations that were previously generated automatically. For example, the perception “I am disliked” may be reconsidered as a possibility rather than a confirmed fact, allowing rigid cognition to loosen. However, this does not always occur; it is a conditional effect that arises only when certain conditions are met.

In this way, NVC does not change the content of dialogue itself, but by structuring the form of expression, it stabilizes the overall quality of communication.

5-3. Limitations (Optimized Version)

NVC is an effective method for dialogue, but it cannot be applied to all situations.

When the other party has a clear aggressive intent, NVC does not function as a means of suppression. In such cases, the very act of receiving a response serves as feedback that the attack is effective, resulting in the amplification and escalation of the behavior.

In contrast, when outputs are generated as a defensive reaction, external intervention is processed as a new form of attack. In this state, requests to adjust expression are interpreted as self-negation, reinforcing opposition rather than reducing it.

The key factor is the updatability of underlying premises.

Even when there is a misalignment in premises, if that misalignment can be shared and reconfigured, NVC can function effectively. However, when premises are fixed and resistant to external revision, this reconfiguration does not occur, and NVC fails to operate.

In this way, NVC is effective when a certain level of dialogue is established and when the underlying premises remain open to updating. When these conditions are not met, it becomes ineffective or may even produce the opposite effect. This does not deny the validity of NVC, but rather defines the conditions under which it can be applied.

6. Positioning of COS (Internal Processing)

6-1. Processing of Premises and Context

Outputs in dialogue may appear to be generated in the moment, but in reality they arise from internal processing of underlying premises and context. COS is a framework that addresses this internal structure.

Here, “premises” refer to the initial conditions that determine how input is interpreted, while “context” refers to the process through which meaning is constructed based on those premises. These processes are not directly observable from the outside; only the resulting output can be observed.

As a result, misalignment in dialogue is often perceived as a problem of output, whereas it frequently originates at the stage of premise and context processing. COS treats this invisible internal layer as its subject, focusing on the structural mechanisms behind the generation of output.

6-2. Mechanism of Misalignment

Misalignment in dialogue does not arise as a single misunderstanding, but as a result of branching and asynchrony within internal processing.

When different premises are applied to the same input, processing diverges into multiple pathways. This branching occurs from the early stage of dialogue, but does not immediately become observable as a problem.

Each diverged pathway maintains its own internal consistency, so the interaction may appear coherent on the surface. At this stage, the misalignment remains latent.

As processing continues, the difference between premises may exceed a certain threshold, at which point the misalignment becomes explicit. This is detected as a mismatch in premises and begins to affect the continuity of dialogue.

It is important to note that the misalignment does not originate at the moment of detection; it has already existed since the initial divergence of premises.

Once the misalignment becomes explicit, the process may be resynchronized if the underlying premises are open to revision. However, if the premises are fixed, the asynchronous state persists, often leading to breakdown or escalation in the interaction.

In this way, misalignment emerges structurally from asynchronous processing caused by divergent premises and its delayed detection.

6-3. Explanatory Capacity of Internal Structure

COS does not directly evaluate outputs in dialogue; instead, it focuses on explaining the underlying internal processing structures that generate them.

Observed statements and behaviors in dialogue are results, and their causes cannot be uniquely determined from the outputs alone. However, through the structural framework of premise and context processing, COS provides a way to explain which processing pathways may have been selected and how they led to a given output.

It is important to note that COS does not produce a single correct interpretation. Since internal processing is not directly observable, any explanation necessarily involves multiple possible pathways.

Therefore, the role of COS is not to determine a definitive interpretation, but to structurally present the possible premise divergences and processing routes that could have generated the observed output.

Through this, misalignment in dialogue can be understood not merely as an error, but as the result of underlying internal processes.

7. What Scope Is (Core Definition)

This chapter presents the core definition of Scope. The descriptions in the previous sections serve as an introduction and overview, and the definition provided here establishes the structural reference framework for this document.

7-1. A Model for Inferring the “Pathways” of Thought Generation

Scope is defined as a model for structurally inferring the pathways of thought generation that lead to observed outputs or fragmentary information.

Statements and actions observed in dialogue are results, and the generative process behind them cannot be directly specified from those results alone. However, behind any output there are a series of processes, including the selection of premises, the application of context, and branching in interpretation, all of which can be understood as a continuous pathway.

Scope does not treat this process of thought generation as a single fixed result. Instead, it treats it as a fluid pathway containing possible branches, and infers its structure from observable fragments.

This model is constructed in dependence on the conditions of observation, and it does not directly describe internal processing itself. While COS is concerned with explaining internal structure, Scope takes that structure as a premise and serves to infer what kinds of pathways may have been selected.

In this way, Scope is positioned between output and internal structure, treating the flow of thought generation as a pathway and making it possible to infer a reconstruction from result back to process.

7-2. A Structure for Maintaining Multiple Pathway Hypotheses

Scope does not determine a single hypothesis for a thought generation pathway; instead, it maintains multiple pathway hypotheses simultaneously.

In dialogue, observed outputs are results, and the underlying internal processing cannot be directly observed. Therefore, it is not possible to uniquely determine which pathway led to a given output.

If interpretation is forced into a single conclusion, alternative possibilities are excluded, creating a risk of fixing understanding on incorrect premises.

Scope addresses this by maintaining multiple pathway hypotheses at the same time. This allows different premises and processing routes to remain available without over-reliance on a specific interpretation.

While it is possible for humans to consider multiple possibilities, maintaining and comparing them imposes a high cognitive load, and there is a natural tendency to converge on a single interpretation.

In contrast, computational systems such as LLMs can maintain and process multiple hypotheses in parallel, enabling the structural capabilities of Scope to be extended at the implementation level.

This structure is not intended to select a single correct answer, but to preserve a range of interpretations and to work with them as needed. When appropriate, it also enables the selective presentation of pathways that are more effective within a given context.

7-3. Non-Deterministic Premise (Non-Finality)

Scope does not assume that a thought generation pathway can be determined as a single, fixed interpretation.

In dialogue, observed outputs are results, and the underlying internal processing cannot be directly observed. Therefore, it is structurally impossible to determine a single pathway as the unique cause of a given output.

If a specific interpretation is adopted as definitive, alternative possibilities are excluded, creating a risk of fixing understanding on incorrect premises.

Scope accepts this constraint and treats thought generation pathways as non-deterministic. This is not an acceptance of ambiguity, but a structural stance that avoids over-specifying conclusions about unobservable internal processes.

Accordingly, non-determinism in Scope is not a matter of choice, but a necessary condition for maintaining multiple pathway hypotheses.

7-4. N=1 High-Resolution Model

Given this non-determinacy, observation of the thought generation process must begin under specific conditions. Scope is defined as a high-resolution N=1 model constructed based on particular observation conditions.

Since the thought generation process cannot be directly observed from the outside, its structure is described in relation to a specific observer and set of conditions. Due to this constraint, high-resolution structural analysis necessarily begins under a single set of conditions.

While general models are typically constructed by extracting commonalities from multiple cases, Scope is established by prioritizing depth over breadth in a single observed instance. This enables detailed handling of thought generation structures, including premise selection, contextual processing, and interpretive branching.

However, this model is not intended for direct generalization in its existing form. Reproducibility in Scope is not defined by identical outputs, but by whether similar structural patterns reoccur under the same conditions.

Therefore, the $N=1$ characteristic is not a limitation, but a prerequisite for observing the thought generation process at high resolution.

8. Separation of Understanding and Demands (Core)

8-1. Demands Based on Lack of Understanding (Conventional)

In dialogue and human relationships, demands directed at others' statements or actions are often made without sufficient understanding of the underlying thought generation process. This tendency is not limited to human-to-human interaction but also appears in interactions with LLMs.

In such cases, only observable outputs—statements or actions—are available, while the premises, context, and interpretive processes that led to them are not considered. As a result, evaluation and reactions are directed solely at the output itself.

Furthermore, not all factors influencing thought generation are externally observable. Cognitive traits, perceptual limitations, physical constraints, situational and environmental conditions, and internal states may not be explicitly reflected in outputs, making it difficult to infer the underlying conditions based on behavior alone.

Interpretations formed under such conditions tend to become fixed as the basis for subsequent cognition. Even when new information is introduced, it does not necessarily lead to revision of underlying premises. Information that does not align with existing interpretations may be treated as exceptions or secondary elements.

Demands formed in such a state are based on a single interpretation derived from observable output and do not account for the thought generation process or unobserved factors. As a result, such demands are limited to superficial corrections of output and may not lead to structural adjustment.

Additionally, demands based on lack of understanding may be perceived by the other party as a denial of their internal state or premises, potentially triggering defensive or aggressive reactions and further reinforcing division.

Thus, demands based on lack of understanding rely on interpretations derived solely from observable outputs, without considering unobserved factors or the difficulty of revising

underlying premises. As a result, misunderstandings tend to persist, and conflict or division becomes fixed.

For example, when a person does not greet someone, it may be interpreted as rudeness, leading to the demand that “you should greet others.” However, in reality, the lack of greeting may result from cognitive constraints or situational factors. In such cases, a demand based solely on observable output, even if intended to correct behavior, may not satisfy the conditions necessary for execution, resulting in a demand that cannot be fulfilled under the given constraints.

8-2. Requests Based on Understanding

Requests directed at others’ statements or actions take on a different nature when they are formed based on an inferred understanding of the underlying thought-generation process.

In such cases, requests are not constructed solely from the observed output itself, but are formed by considering part or multiple possible pathways of premises, context, and interpretation that led to that output. As a result, the target of the request shifts from simple behavioral correction to adjustments of conditions, environments, or even the form of interaction itself.

Requests based on understanding do not aim to directly correct the other’s output. Instead, they assume the structure that generated the output and select modifications that are feasible within that structure. Consequently, requests shift from directives such as “what should be done” to forms such as “under what conditions it becomes possible.”

Without this structure, requests tend to be formed based solely on observed outputs, often seeking superficial corrections without considering the other party’s conditions or constraints. As a result, when unmodifiable factors such as environmental conditions, physical constraints, or cognitive limitations are involved, the request itself may become impossible to execute. In contrast, requests that incorporate multiple pathway hypotheses can shift from mere behavioral correction to adjustments of conditions or interaction formats.

Furthermore, such requests are more likely to be recognized not as a denial of the other’s internal state or constraints, but as adjustments that take them into account. This tends to reduce defensive or aggressive reactions and helps preserve the possibility of continued dialogue and adjustment.

An important point here is that human cognition tends to converge toward a single interpretation of observed outputs and struggles to maintain multiple pathway hypotheses simultaneously. As a result, interpretations may become fixed before requests based on understanding can even be formed.

On the other hand, LLMs can simultaneously expand multiple premises and contexts and present different pathway hypotheses in parallel. This allows them to provide auxiliary perspectives or possibilities that the user may not have recognized. However, since LLMs

ultimately produce a single output, the presented interpretation may still become fixed in practice.

In this context, the role of an LLM is not to determine a single correct interpretation, but to present multiple candidate interpretations while maintaining them in parallel, thereby supporting the formation of requests based on understanding. This is not merely generating responses, but expanding the range of possible interpretations.

However, even requests based on understanding cannot fully capture all underlying factors. As assumed by Scope, thought-generation processes are always treated as multiple pathway hypotheses and inherently include uncertainty. Therefore, such requests remain provisional and are not fixed.

What matters in this state is not the accuracy of the request itself, but the ability to maintain the possibility of updating premises and conditions. Requests based on understanding function as a means to sustain interaction while preserving adjustability, rather than collapsing into a single fixed interpretation. In this sense, they act as a mechanism to delay fixation into a single interpretation.

8-3. Handling Outputs as Defensive Responses

The statements and behaviors of others are not necessarily generated solely through intentional choices or expressions of personality. Under certain conditions, outputs may be generated as defensive responses aimed at maintaining the stability of internal structures.

In such cases, in some instances, observed aggressive or extreme expressions may not be direct representations of evaluation or intent toward others, but rather outputs generated to cope with discrepancies between external conditions and internal instability. In this sense, outputs function not only as actions directed toward others, but also as mechanisms for maintaining one's internal state.

When evaluation or requests for correction are directed solely at the output, such requests may be perceived as interventions targeting the defensive structure itself. As a result, instead of being suppressed, the output may become further reinforced.

For example, when corrections are requested for aggressive expressions, such requests can function as additional external pressure, potentially increasing internal instability. In such cases, the intensity or extremity of the output may increase as a means of maintaining the defensive structure.

In these situations, what is important is not to directly control the content of the output, but to infer the conditions under which the output is generated. That is, outputs should be treated not merely as statements, but as structural responses generated within the relationship between internal states and external environments.

Additionally, outputs generated as defensive responses are not always consciously selected, and may instead arise as automatic responses aimed at maintaining internal coherence.

Therefore, directly requesting modification of such outputs may not constitute a feasible request.

In such cases, what is required is not the correction of the output itself, but an understanding of the premises and conditions that necessitate the defensive response. Without accounting for the underlying structure behind the output, intervention may not resolve the defensive reaction, but rather reinforce it.

For this reason, when dealing with outputs generated as defensive responses, meaningful change is unlikely to occur unless interventions involve adjustments to underlying premises or conditions, rather than direct modification of the output itself.

However, this understanding does not justify the content or impact of the output, and is intended solely to grasp the structure by which it is generated.

8-4. Understanding Does Not Require Acceptance

Even if one understands the generative structure behind another person's statements or actions, there is no obligation to accept their content or demands.

Understanding involves inferring the premises, context, and internal states underlying an output and holding them as multiple pathway hypotheses. It does not, in itself, imply agreement or approval. Therefore, responses based on understanding do not necessarily entail acceptance or compliance.

In particular, even when an output is generated as a defensive response, if its content imposes harm or burden on others, rejecting it constitutes a structurally valid and legitimate claim. The fact that it is defensive does not nullify its impact or consequences.

For this reason, within interactions based on understanding, it is necessary to separate "why the output was generated" from "whether to accept it." The former concerns structural comprehension, while the latter is a matter of choice.

Rejection does not necessarily imply conflict or the breakdown of a relationship. It can function as part of resetting conditions or adjusting the form of interaction (e.g., limiting contact frequency, redefining roles). In this sense, rejection grounded in understanding is not mere opposition, but can serve as a means of maintaining or restructuring a relationship.

However, engagement based on understanding does not always lead to stabilization. Criticism or rejection that directly targets another's underlying premises or internal structure can act more strongly than surface-level disagreement, potentially destabilizing internal coherence. As a result, such interventions may not reinforce defensive structures, but instead contribute to their collapse, which can lead to relational breakdown or the fixation of conflict.

Thus, while understanding increases the precision of intervention, it also increases its potential impact. Engagement based on understanding is not merely an act of empathy or consideration, but can function as a strong form of intervention depending on the situation.

What is important is that understanding does not constrain available options, but rather expands them. It does not compel acceptance, but serves as a foundation for selecting among multiple possible responses.

9. Intervention Boundaries (Application)

9-1. Why Definitive Conclusions Cannot Be Made

The thoughts and intentions of others cannot be directly observed from the outside. In both humans and LLMs, what is observable is limited to outputs such as language and behavior, while the underlying premises, context, and interpretive processes cannot be fully accessed externally.

Scope operates on this fundamental limitation and constructs multiple pathway hypotheses from observed outputs. Therefore, the interpretations it provides are not definitive answers, but sets of inferences dependent on observation conditions.

Given this structure, no interpretation can be determined with certainty. Even if one interpretation appears strongly supported, this does not imply that other possibilities have been eliminated; it merely indicates that, within the observed context, it has relatively higher consistency. While some pathway hypotheses may later turn out to align with actual conditions, such alignment is only identifiable retrospectively and cannot be determined in advance.

Moreover, factors influencing thought generation include cognitive traits, perceptual constraints, physical limitations, environmental conditions, and internal states—many of which cannot be directly observed. These factors may not be explicitly reflected in outputs, making it impossible to uniquely determine internal states based solely on observable information.

For example, even a simple action such as “going for a walk” may not be driven by a single motive. It may involve multiple simultaneous factors such as exercise, mood regulation, incidental purposes, or curiosity about a new place. In some cases, the action may be taken without a clearly articulated reason. Thus, observable behavior is not necessarily generated from a single cause or clearly defined intention.

For this reason, interpretations within Scope must always be treated as provisional rather than fixed conclusions. Definitive interpretations collapse multiple pathway hypotheses into a single assumption, ignoring the inherent multiplicity of possible generative processes.

Furthermore, when an interpretation is presented as definitive, it may function not as understanding but as external labeling. In such cases, interaction shifts from adjustment to fixation, reinforcing division. This can lead to the stabilization of incorrect premises and further reduce the possibility of mutual understanding.

Therefore, interpretations in Scope should be maintained as updateable inferential states rather than fixed judgments. What matters is not the certainty of correctness, but the ability to engage while preserving multiple possibilities.

9-2. Engagement in an Inferred State

Engagement within Scope does not rely on definitive understanding, but operates while maintaining multiple pathway hypotheses in an inferred state.

Rather than intervening based on a single interpretation assumed to be correct, this approach sustains interaction while preserving multiple possibilities. In this context, engagement is not treated as fixed based on a single premise, but as adaptive and continuously updated according to the situation.

In engagement under an inferred state, the priority is not the interpretation itself, but maintaining the ability to update interpretations. Even when responding based on a given hypothesis, it is not treated as the sole premise, and the existence of alternative possibilities is preserved.

The hypotheses being held may include incorrect ones and can be discarded during the process of engagement. Therefore, instead of adhering to a specific hypothesis, ongoing revision and switching of hypotheses occurs in response to changing conditions.

When premises and conditions are uncertain, this type of engagement may appear ambiguous or inconsistent from an external perspective. At the same time, however, behavioral expectations and boundary settings are maintained clearly.

Such engagement does not provide definitive conclusions and therefore includes uncertainty. However, this uncertainty is not a flaw, but a necessary condition for maintaining multiple pathway hypotheses. Attempting to eliminate uncertainty results in convergence toward a single interpretation.

Furthermore, engagement in an inferred state prioritizes adaptability over strict consistency. Even when responses based on different hypotheses are presented over time, this should not be considered contradiction, but rather updates reflecting changing observations and new information.

Thus, hypothesis revision and switching may continuously occur throughout the engagement process. What matters is not maintaining a particular interpretation, but preserving the ability to update appropriately as conditions change.

Engagement in an inferred state is not intended to produce a single conclusion, but to facilitate ongoing adjustment while maintaining multiple possibilities within interaction. Through this structure, fixation of interpretation is prevented, reducing the likelihood of division and its reinforcement.

9-3. Non-Intervention / Translation / Adjustment / Control

Engagement in an inferred state does not apply a single type of intervention to all situations; rather, it requires selecting the intensity and form of engagement depending on the context. Within Scope, engagement can be broadly categorized into four modes: non-intervention, translation, adjustment, and control.

Non-intervention is the choice to observe the progression of a situation without applying structural interpretation or intervention. Immediate intervention is not always appropriate for every output or interaction, and prioritizing observation can enable more accurate judgment. In non-intervention, approaches such as Nonviolent Communication (NVC) serve as effective options. By organizing emotions and needs, they help stabilize interaction without directly intervening in underlying structures. However, these methods do not directly intervene in the generative structure of the output, and depending on the situation, may need to be combined with other forms of engagement.

Translation is a form of engagement that preserves the content of the output while transforming its interpretation or expression. For example, rather than accepting aggressive language at face value, one may infer the underlying intent or state and reconstruct it in a way that can be communicated to others. The goal is not to alter the content itself, but to increase the transmissibility of its meaning. While interpretive support is provided, this does not include distortion of the original intent.

Adjustment targets not the output itself, but the conditions and relationships that produce it. For instance, modifying interaction frequency, distance, roles, or environment can alter the conditions under which outputs arise. This approach induces structural change without directly correcting the output.

Control involves directly imposing constraints or requesting changes on outputs or behaviors. It represents the strongest form of intervention and is selected when other methods are insufficient or when the impact is significant. While control can produce immediate effects, it also carries a high risk of triggering resistance or defensive reactions. Therefore, control should be limited to situations where the impact or burden on others is substantial.

These modes of engagement are not used in a fixed manner, but shift dynamically depending on the situation. For example, early stages may involve non-intervention or translation, followed by adjustment or control as needed. This selection is based on factors such as impact, urgency, and the stability of the interaction.

What matters is not that any particular mode is always correct, but that the appropriate level and form of engagement is selected according to the situation. Engagement in an inferred state requires not reliance on a single method, but the dynamic switching between multiple modes.

At present, these modes of engagement and their switching conditions are based on limited observation (N=1), and broader validation and professional design are required to ensure stable application across diverse conditions.

9-4. Responses That Do Not Fixate Division

Division arises when a single interpretation becomes fixed and its underlying premises are no longer updated. When a specific meaning or intention is definitively assigned to an output, it functions as a label imposed on the subject, constraining subsequent interaction.

Such fixation not only makes mutual understanding difficult, but also reinforces and perpetuates conflict. Once an interpretation is formed, it becomes resistant to change even when new information is introduced, resulting in a structure where division persists.

Responses within Scope aim to prevent this fixation. Rather than converging on a single interpretation, they maintain multiple pathway hypotheses, preserving the possibility of updating underlying premises.

The key point here is not to maintain ambiguity, but to delay the fixation of interpretation. By holding multiple possibilities simultaneously, interpretations remain open and continue to evolve through interaction.

Responses that do not fixate division are not intended to eliminate all conflict. When understanding or adjustment is not possible, setting boundaries or rejecting interaction may be necessary. However, even in such cases, avoiding reduction of the other's output or intent to a single label helps prevent the fixation of division.

This does not deny division itself, but aims to prevent its fixation.

Therefore, responses within Scope are not primarily about achieving agreement or resolving conflict, but about maintaining the possibility of updating interpretations. Through this structure, division is not fixed, but treated as a variable state.

This is not an avoidance of judgment, but a structure that prevents the fixation of the premises underlying judgment.

9-5. Activation Conditions and Operational Cost (Implementation Considerations)

In the operation of Scope, applying it continuously to all outputs is not realistic. The process of generating and maintaining multiple pathway hypotheses incurs significant computational cost, potentially increasing response latency and system load. Therefore, rather than being constantly active, Scope is better designed to be triggered under specific conditions.

One example of activation conditions is when fixation of interpretation is likely occurring. This includes statements that reduce causes to a single explanation, such as “Why is that person always like this?” or “Why am I always like this?”, as well as strong assertions or extreme evaluations. In such cases, presenting multiple pathway hypotheses can indicate that alternative possibilities may exist beyond the fixed premise.

Additionally, when decision-making appears to have converged excessively into a single option, Scope intervention becomes particularly relevant. By introducing the possibility of multiple factors and pathways, it may create fluctuation against convergence toward a single conclusion and allow cognitive re-expansion.

Furthermore, when convergence toward extreme conclusions or choices extends into domains related to life or safety, especially careful design is required. In such states, judgments based on a single premise tend to be reinforced, and the outcome of interaction may vary significantly depending on the intensity and timing of intervention. Therefore, activation conditions and response design in these cases require a different level of validation, including specialized verification, design, and environment, compared to general interactions.

The design of such activation conditions may be achievable through relatively simple keyword-based triggers; however, determining appropriate conditions is currently based on limited observations. Broader applicability requires further validation and design. In particular, preventing excessive activation or false positives necessitates adjustments tailored to the operational environment.

Moreover, Scope is hypothesized to be implementable as an external layer without requiring major modifications to the internal structure of existing LLMs. In other words, by adding interpretive support and hypothesis generation as external processes, a staged implementation approach can be envisioned. Similarly, certain functions of COS may also be partially implemented externally, although the details of such implementation require further investigation.

Therefore, the implementation of Scope should not assume a single finalized form, but rather be designed incrementally according to operational conditions and environment. The key is not the complete reproduction of the theory, but its adaptation into a practically applicable form that balances cost and effectiveness.

10. Summary

10-1. Moving Beyond Single Interpretation

Human and LLM thought generation inherently allow for multiple possible pathways, yet in practice, outputs tend to converge into a single interpretation or conclusion. While this convergence contributes to efficiency and rapid decision-making, it can also lead to the omission of underlying premises and context, resulting in fixed interpretations and persistent misunderstandings.

Convergence to a single interpretation is a natural mechanism for reducing cognitive load. However, in this process, alternative possibilities become invisible. As a result, misalignment is more likely to occur between individuals who operate under different contexts, premises, environmental conditions, cognitive traits, perceptual constraints, physical limitations, internal states, and temporal conditions, contributing to division in dialogue and interaction.

What is important here is not the complete elimination of single interpretations, but rather treating them as selectively adopted when necessary. In some situations, arriving at a single conclusion is required, and this selection itself is not problematic. The issue arises when that conclusion becomes fixed as the only possible one, excluding all alternatives.

Therefore, moving beyond single interpretation does not mean abandoning conclusions, but engaging while maintaining the possibility that multiple interpretations may exist behind any given conclusion. In this structure, interpretations are not fixed but remain open to revision depending on context.

This does not avoid decision-making, but rather prevents the fixation of the premises upon which decisions are made.

10-2. The Importance of Maintaining an Inference-Capable State

The core of Scope, as discussed in this work, lies in maintaining an “inference-capable state.” This refers to a condition in which multiple pathway hypotheses are preserved and remain updateable, rather than converging into a single interpretation.

In such a state, interpretations of outputs are not fixed and can be revised in response to new information or changes in observation conditions. As a result, even when misunderstandings or misalignments occur, there remains room for adjustment through updating underlying premises.

Conversely, when the inference-capable state is lost, interpretation converges into a single premise, excluding alternative possibilities. In this condition, interaction loses flexibility and shifts into a structure where misunderstandings and conflicts become difficult to resolve.

In such states, judgments based on incorrect premises are more likely to persist, making correction through dialogue increasingly difficult. This can result in the fixation of misunderstanding, the continuation of unnecessary conflict, or inappropriate decision-making. These issues extend beyond interpersonal relationships and can affect quality and safety in system operation and service delivery.

Furthermore, responses based on single interpretations may, particularly in systems designed to prioritize empathy, reinforce the user’s premises excessively. For example, when a justification for a specific behavior is accepted without examination, the user may interpret this as validation, reinforcing their belief that their judgment is correct. In such

cases, the structure does not facilitate mutual understanding, but instead strengthens flawed premises and interpretations.

An inference-capable state does not simply mean having multiple interpretations. The essential point is that these interpretations are held simultaneously and remain updateable according to the situation. This capacity for revision forms the foundation for flexibility and adjustability in interaction.

Although this state includes uncertainty, it is not a defect but a necessary condition for maintaining multiple pathway hypotheses. Attempts to eliminate uncertainty tend to result in convergence toward a single interpretation.

Therefore, maintaining an inference-capable state functions as a foundational condition for correcting misunderstandings, mitigating conflict, and preventing the fixation of division. Scope serves as the structural framework that sustains this state.

10-3. Role Separation of Scope / COS / NVC

Scope, COS, and NVC, as discussed in this work, each serve distinct roles and are not alternative solutions to the same problem. Rather than competing, they function as complementary components operating at different layers.

Scope serves as a structural foundation for inferring pathways of thought generation and maintaining multiple pathway hypotheses. It does not evaluate the correctness of outputs or conclusions, but instead handles premises, context, and interpretive branches to prevent fixation on a single interpretation.

COS addresses the structure of internal cognitive processing, describing a sequence of processes including premise handling, contextual interpretation, and output generation. While Scope operates as an external inferential model, COS represents a framework for internal processing.

NVC is a method focused on expression and communication in interaction, utilizing elements such as observation, emotion, needs, and requests to enhance relational stability. It primarily operates at the level of output form and relational adjustment.

These roles do not overlap but are positioned across different layers. Scope is responsible for structural understanding and maintaining non-fixation, COS for describing and regulating internal processing, and NVC for expression and communication in interaction.

Therefore, Scope does not replace NVC, nor is COS subsumed within Scope. Each serves a distinct function, and by being combined as needed, they enable a comprehensive approach to thought generation and interaction.

This separation of roles prevents the conflation of structure, processing, and expression, allowing for the appropriate application of methods within each domain.

Through this role separation, LLMs can suppress convergence to single interpretations and excessive fixation on premises, enabling more flexible and adjustable response generation.

10-4. Implementation Conditions

The application of Scope is not intended to be continuous across all situations, but rather to be triggered under specific conditions. The process of generating and maintaining multiple pathway hypotheses incurs significant computational cost, affecting response speed and system load. Therefore, it is preferable to apply Scope selectively in situations where fixation of interpretation or convergence to a single pathway is likely occurring.

Such activation is assumed to be designed based on specific speech patterns or keywords; however, appropriate condition setting is currently based on limited observations, and broader applicability requires further validation and adjustment.

Moreover, Scope is considered to be implementable as an external layer without requiring major modifications to the internal structure of existing LLMs. This enables a staged or partial integration into existing systems.

On the other hand, the design and validation of these mechanisms require not only technical considerations but also a stable environment and continuous evaluation conditions. Particularly when dealing with high-density cognitive processing, environmental factors become elements that influence design precision.

Therefore, the implementation of Scope should not assume a single finalized form, but should be designed and applied incrementally according to operational conditions and environment. The key is not the complete reproduction of the theory, but its adaptation into a practically viable form that balances cost and effectiveness.

Appendix

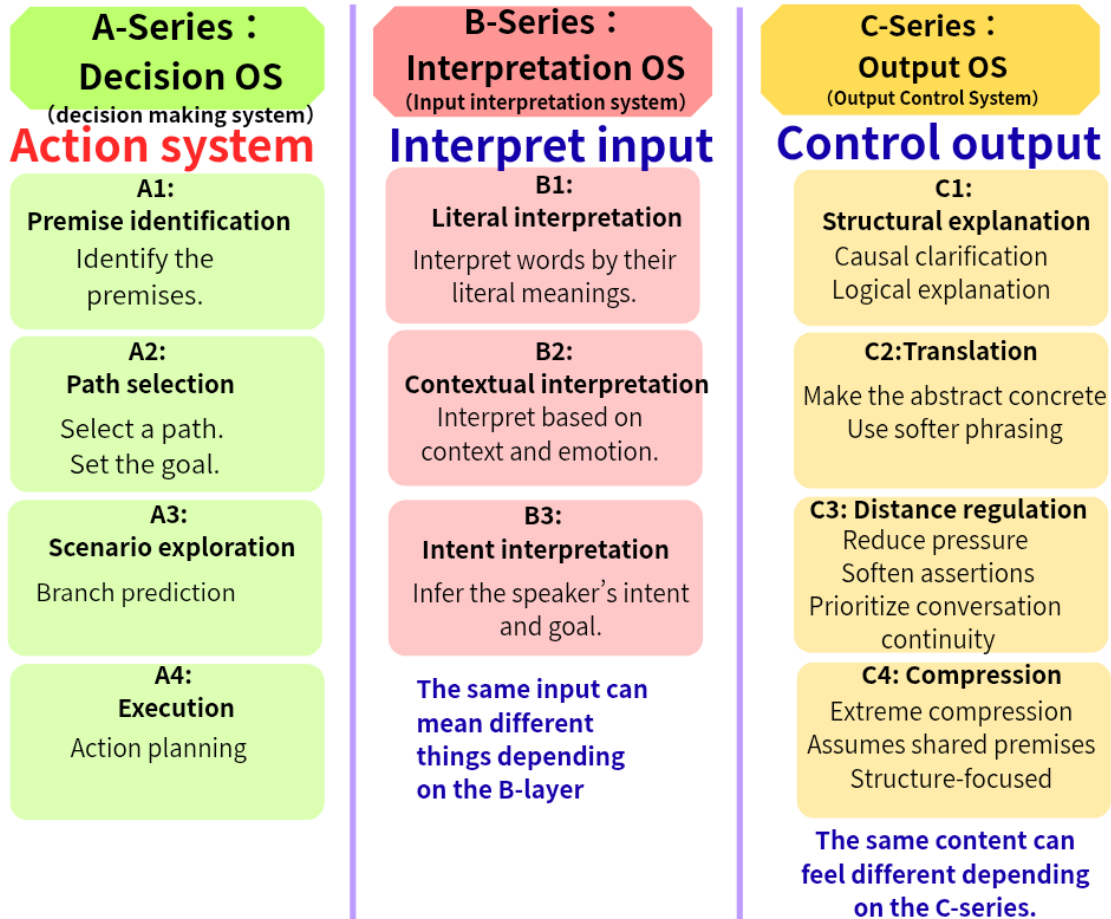
Appendix: Overview Diagram of the Internal Processing Structure (COS)

This diagram provides a high-level overview of the structure. For detailed definitions of each component, please refer to the separate COS document.

COS

Cognitive Operating System

Scope detects layer errors in thought, while COS detects premise misalignment. Both operate in parallel.



D-Series : Internal /Structural Engine

(Internal Structure System)

Analyze structure

Monitoring loop (Human) Verification loop (LLM)

